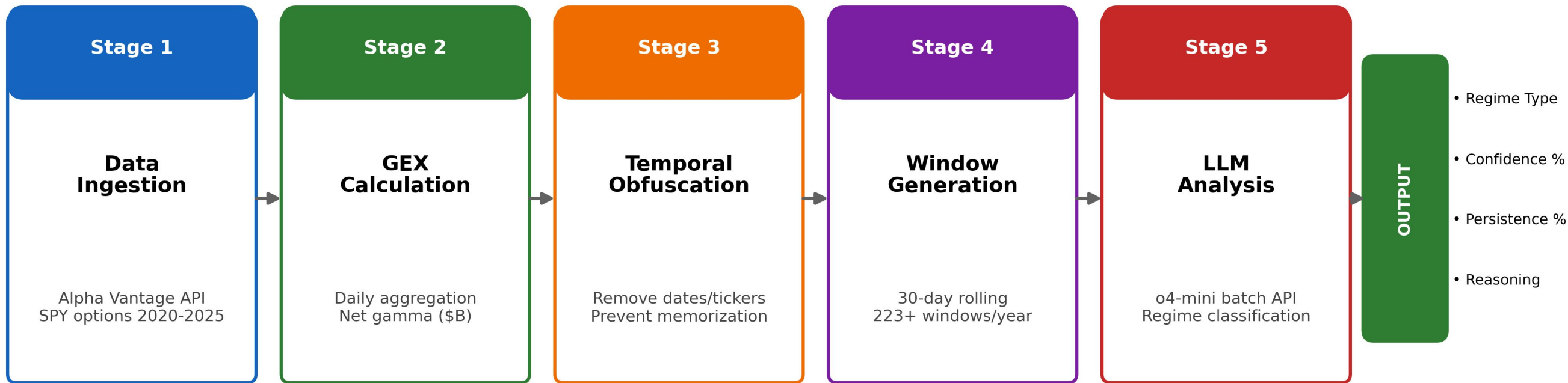
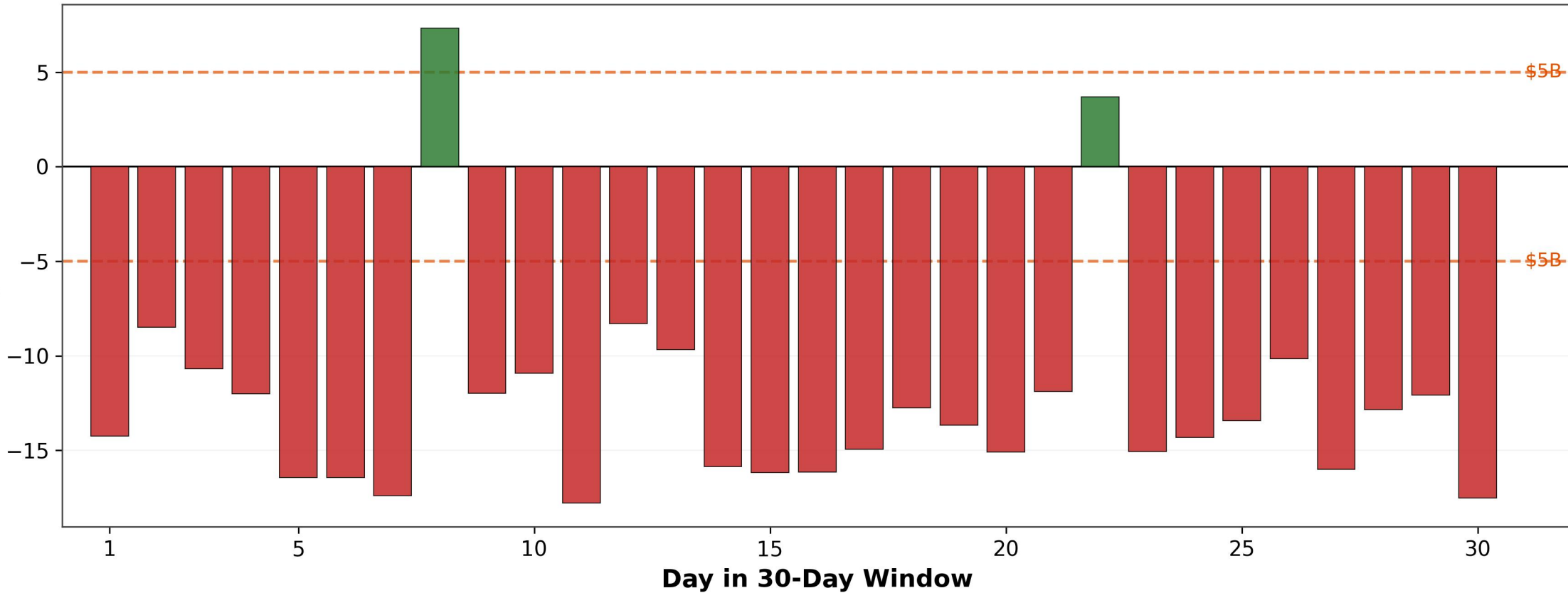




GEX Magnitude (\$B)



Persistence: 28/30 (93.3%)

Avg Magnitude: \$13.1B

Sign Flips: 2

Classification: PERSISTENT NEGATIVE

Temporal Obfuscation Process

Preventing LLM Memorization While Preserving Structural Information

BEFORE

Original Data

Date:	2024-03-15
Ticker:	SPY
Day Type:	Friday
GEX:	-\$12.3B
Persistence:	93%
Sign Flips:	2

OBFUSCATION

- Remove calendar dates
- Strip ticker symbols
- Remove day-of-week
- Preserve magnitudes
- Keep structure intact

AFTER

Obfuscated Data

Date:	Day 15
Ticker:	[REDACTED]
Day Type:	[REMOVED]
GEX:	-\$12.3B
Persistence:	93%
Sign Flips:	2

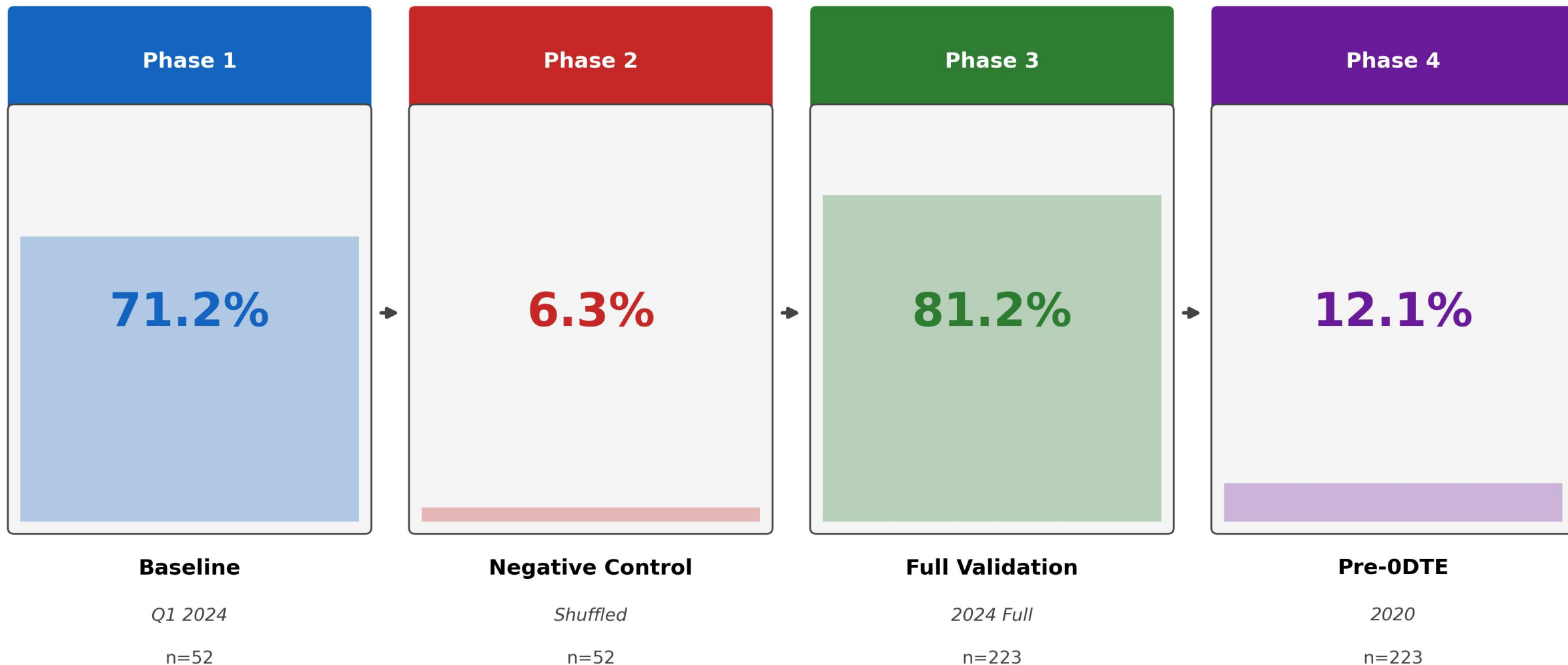


REMOVED: Temporal identifiers that could enable memorization

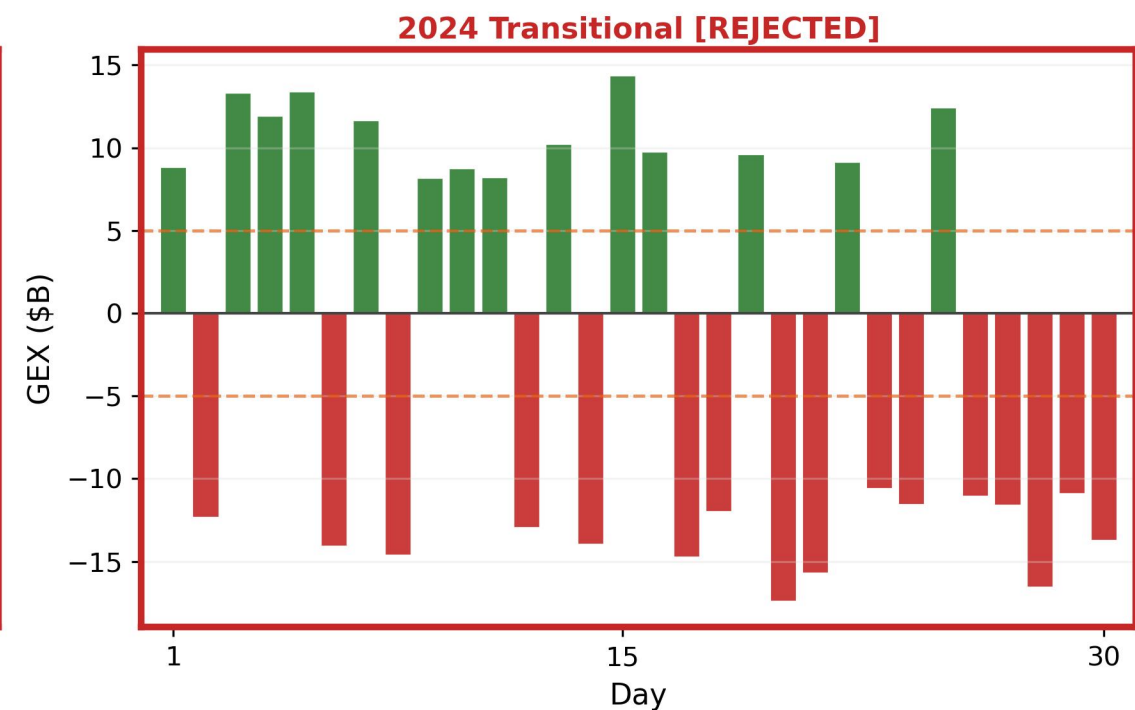
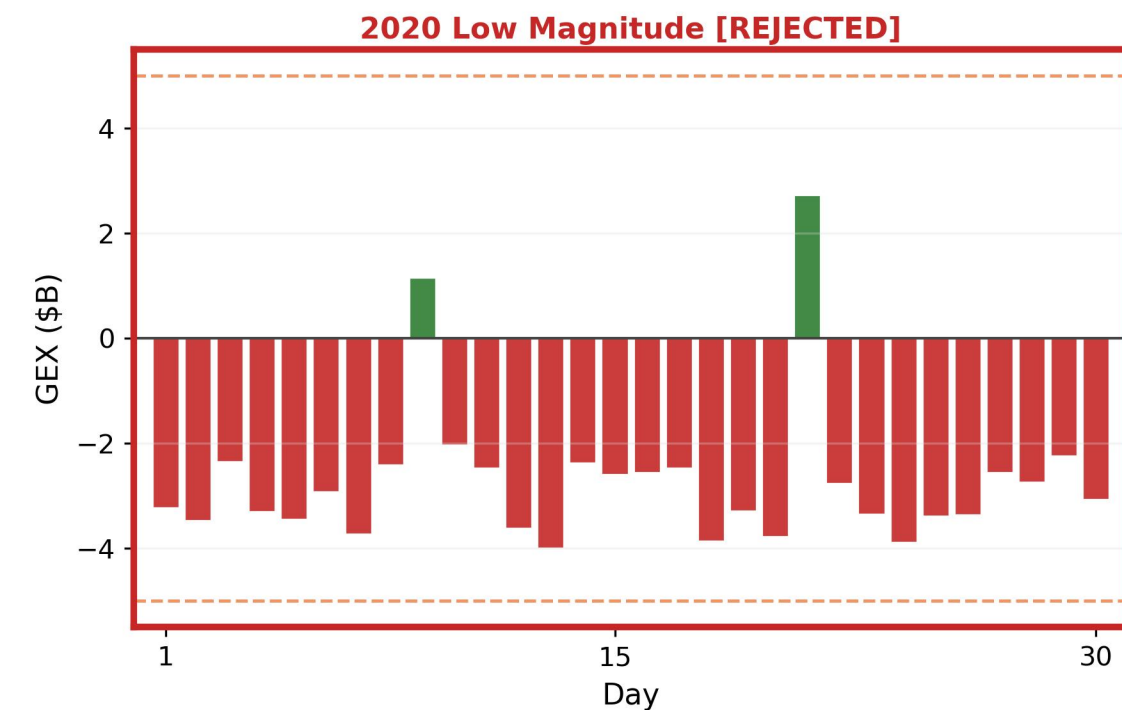
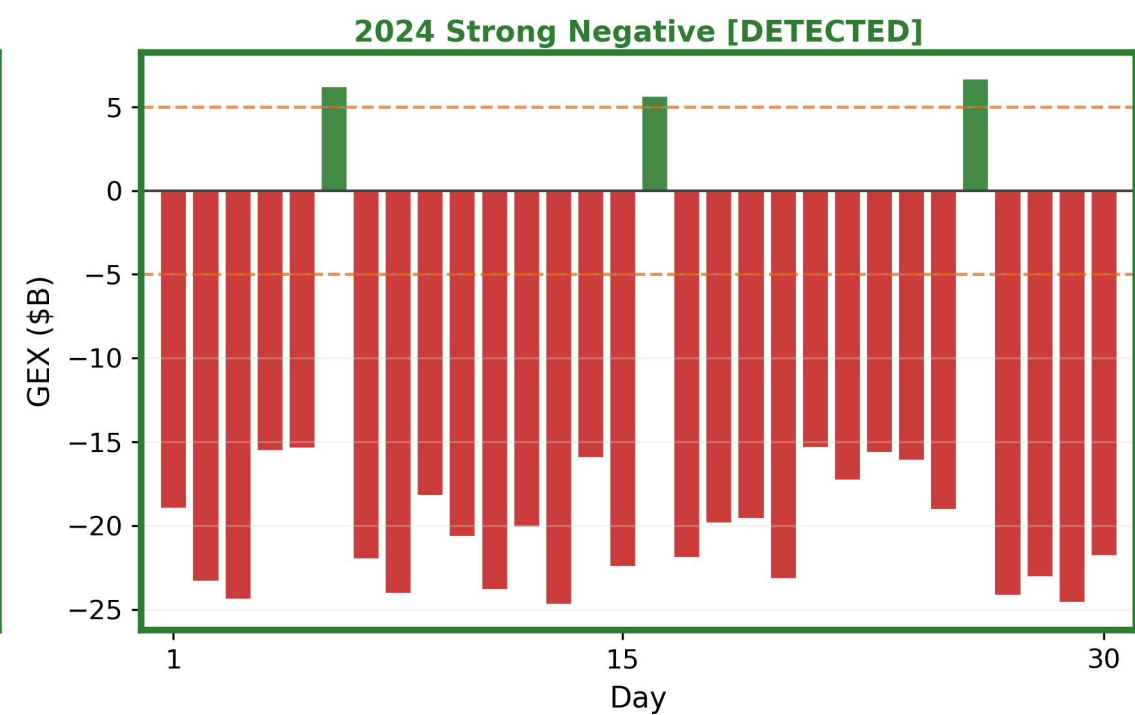
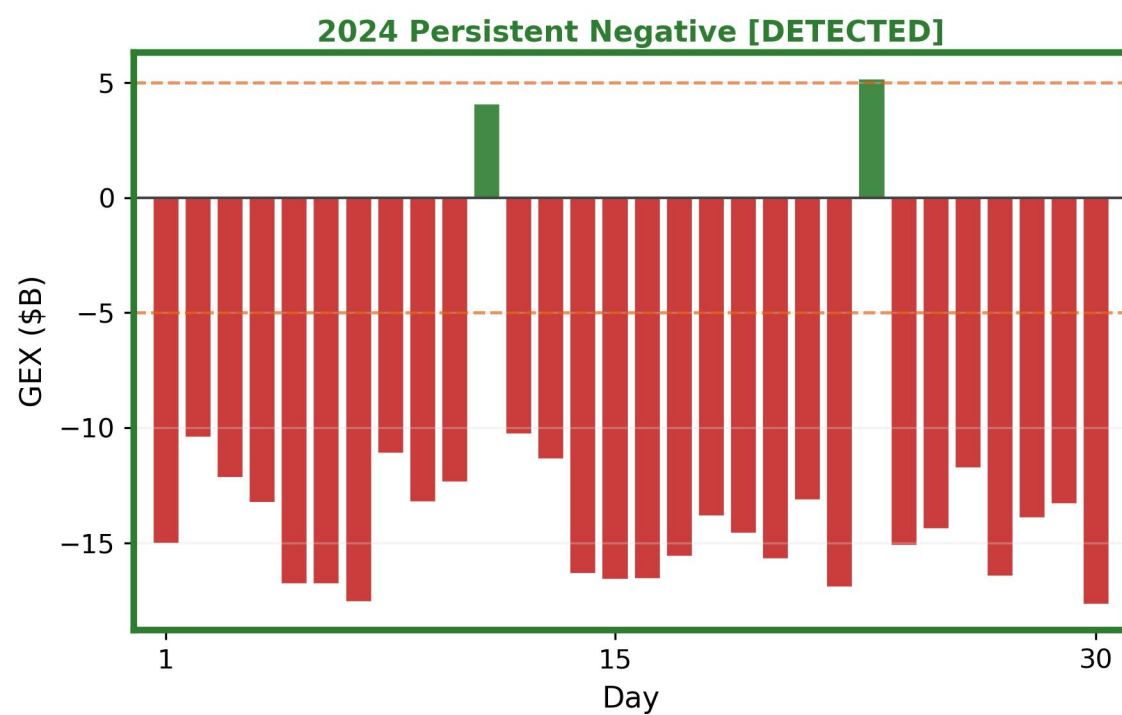


PRESERVED: Structural metrics required for regime detection

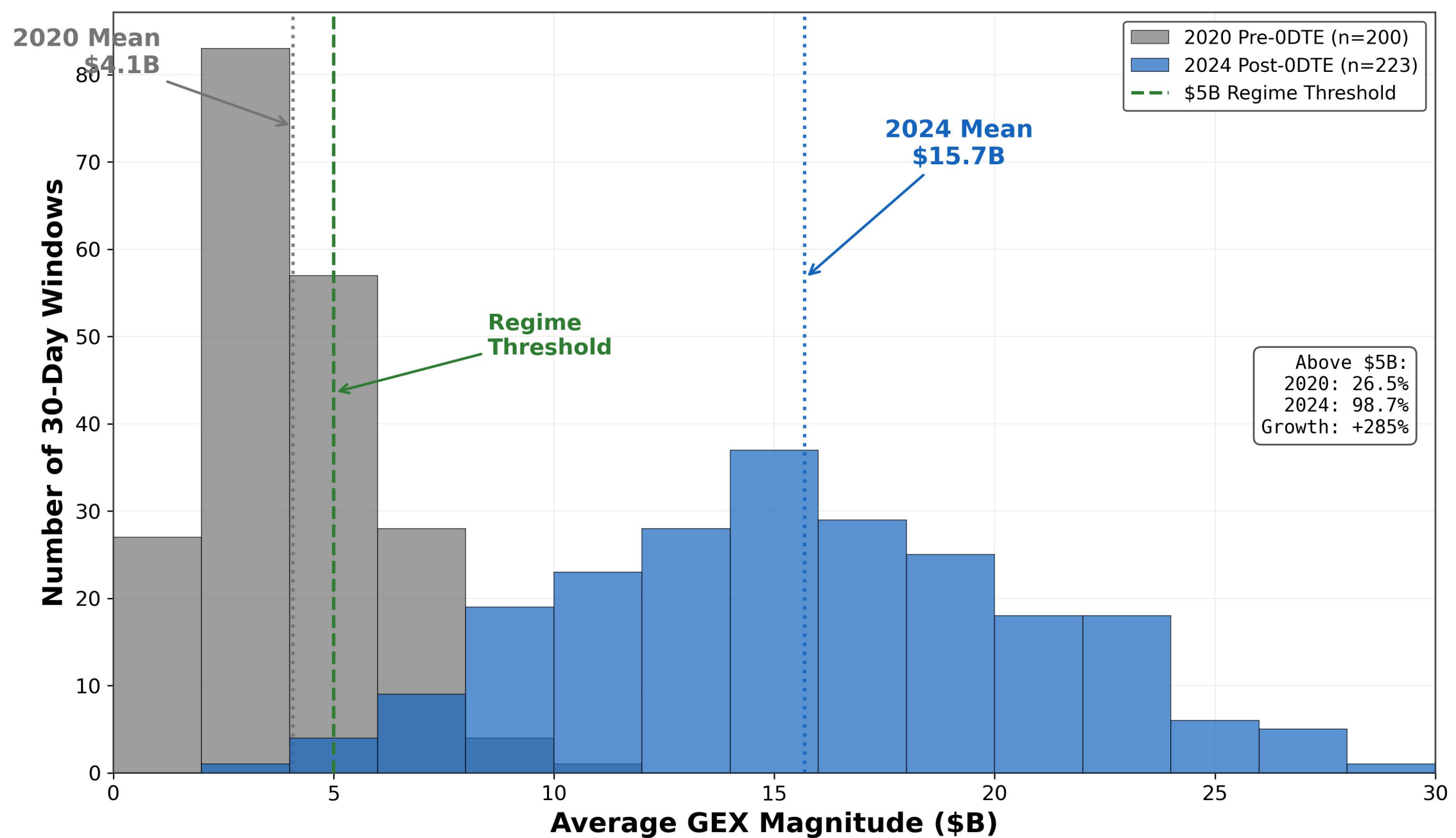
Why Obfuscation Matters: LLMs trained on financial data could memorize that "March 2024 had negative gamma" without understanding the mechanics. By removing temporal anchors, we force the model to reason from structural patterns (persistence, magnitude, sign flips) rather than recalling specific dates.



Key Result: 69.1pp discrimination between 2024 (81.2%) and 2020 (12.1%) • $p < 0.0001$ • $\phi = 0.672$

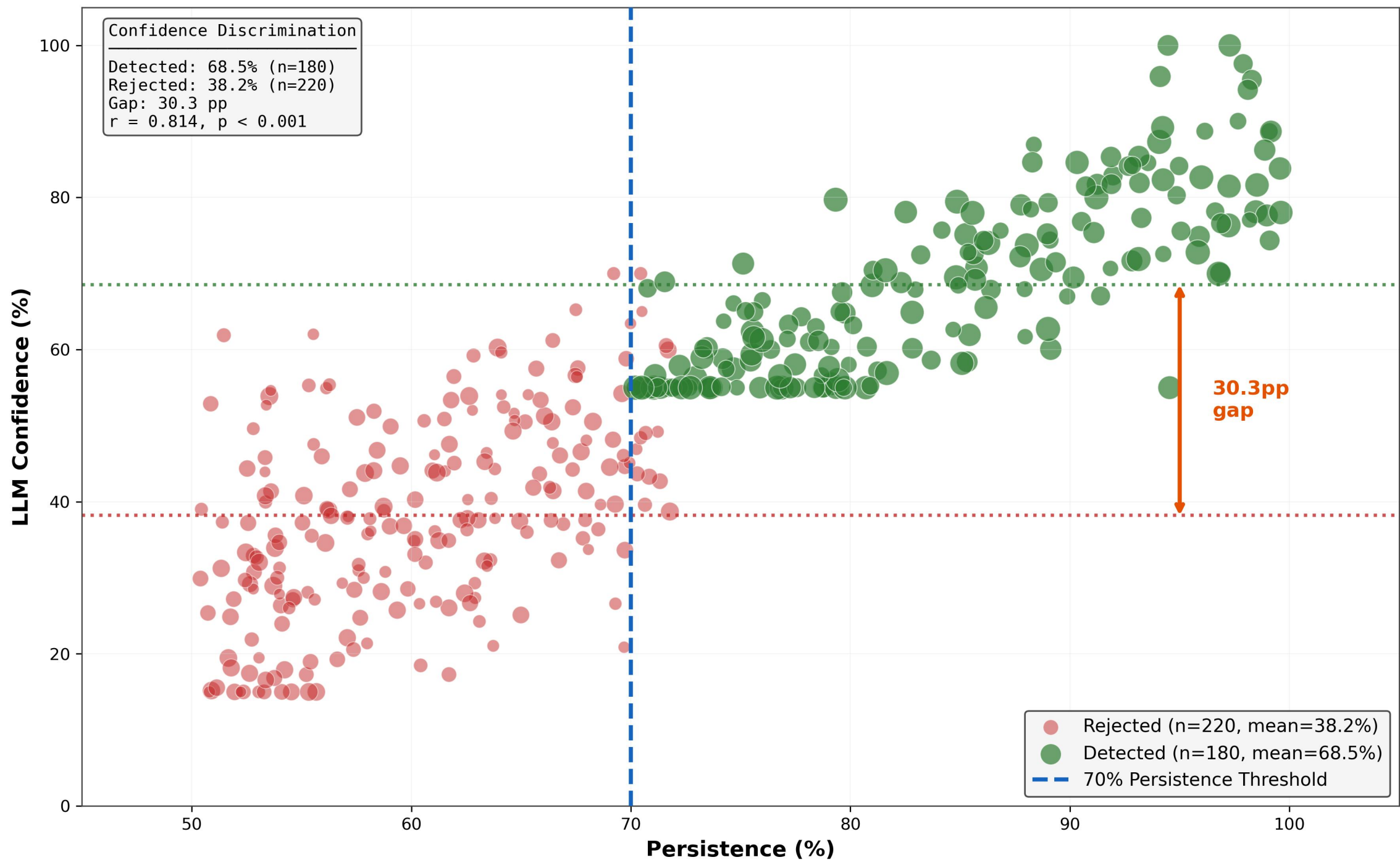


Detection Criteria: Persistence > 70% | Avg Magnitude > \$5B | Sign Flips ≤ 5

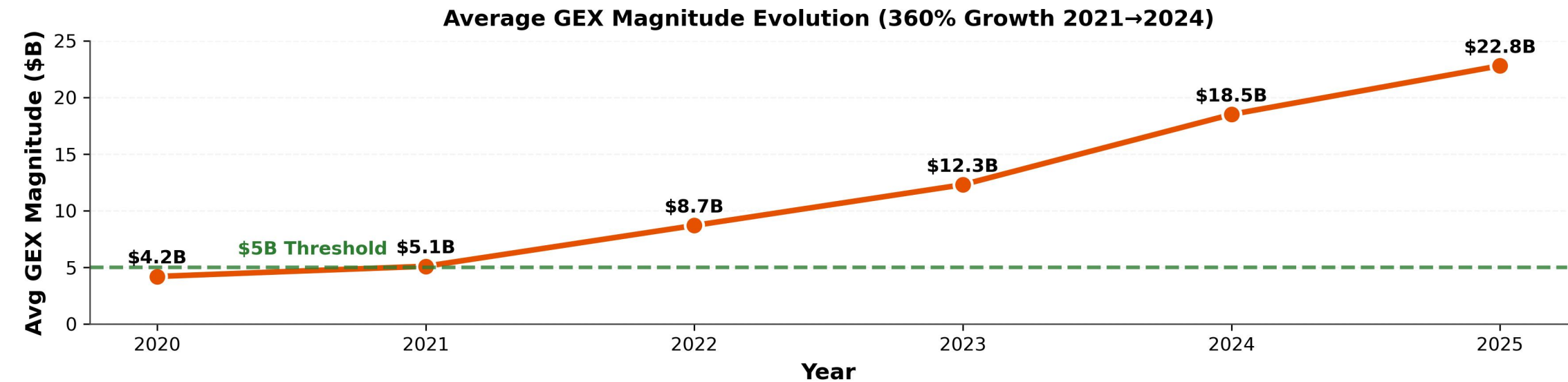
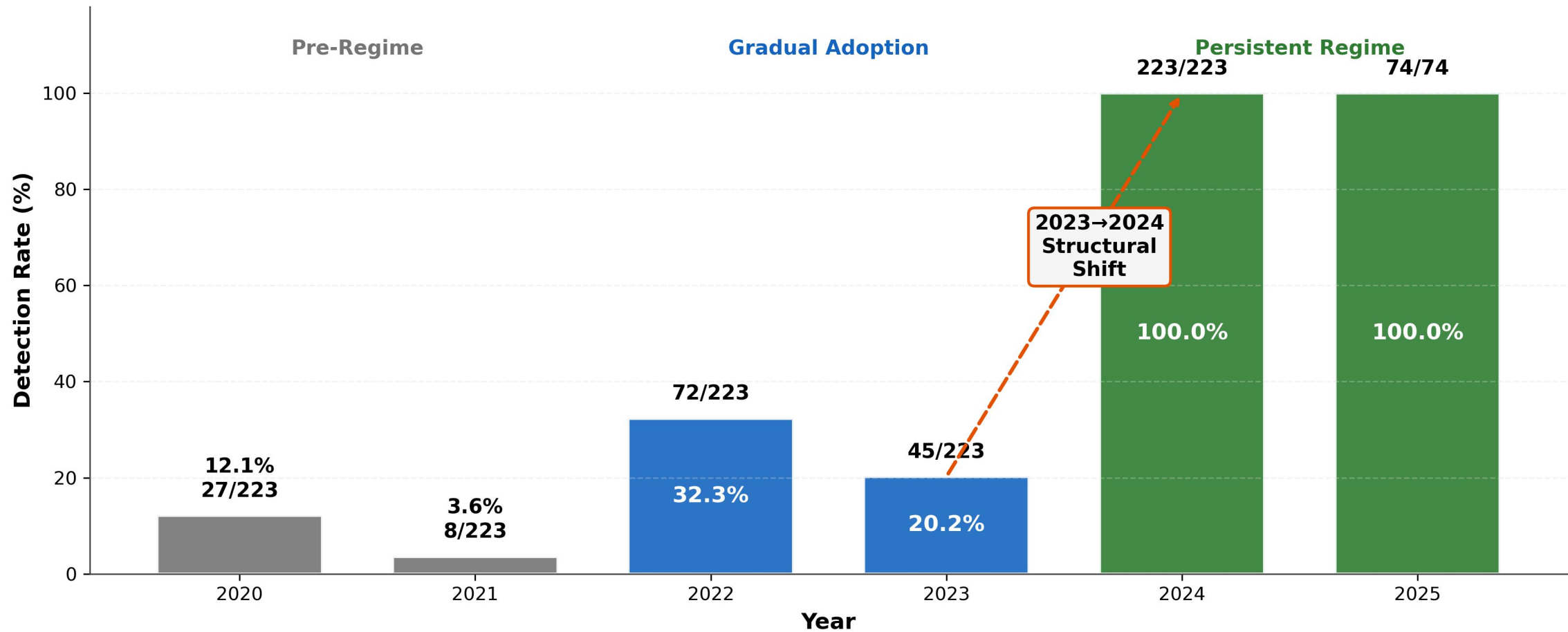


LLM Confidence Discrimination Across Full Detection Spectrum

Point size encodes GEX magnitude (larger = higher dealer positioning)



Phase 4A: Temporal Progression of Regime Detection (2020-2025) Gradual ODTE Adoption with 2023→2024 Structural Market Shift



Key Finding: Detection rates track market evolution precisely. Low rates in 2020-2021 (12.2%, 3.7%) reflect pre-regime baseline. Growing but inconsistent rates in 2022-2023 (32.4%, 20.2%) show gradual ODTE adoption. Perfect 100% detection in 2024-2025 marks structural shift with sustained dealer gamma regimes. GEX magnitude grew 360% (5B→23B), far exceeding inflation (20-25%).

INTRADAY MECHANISM

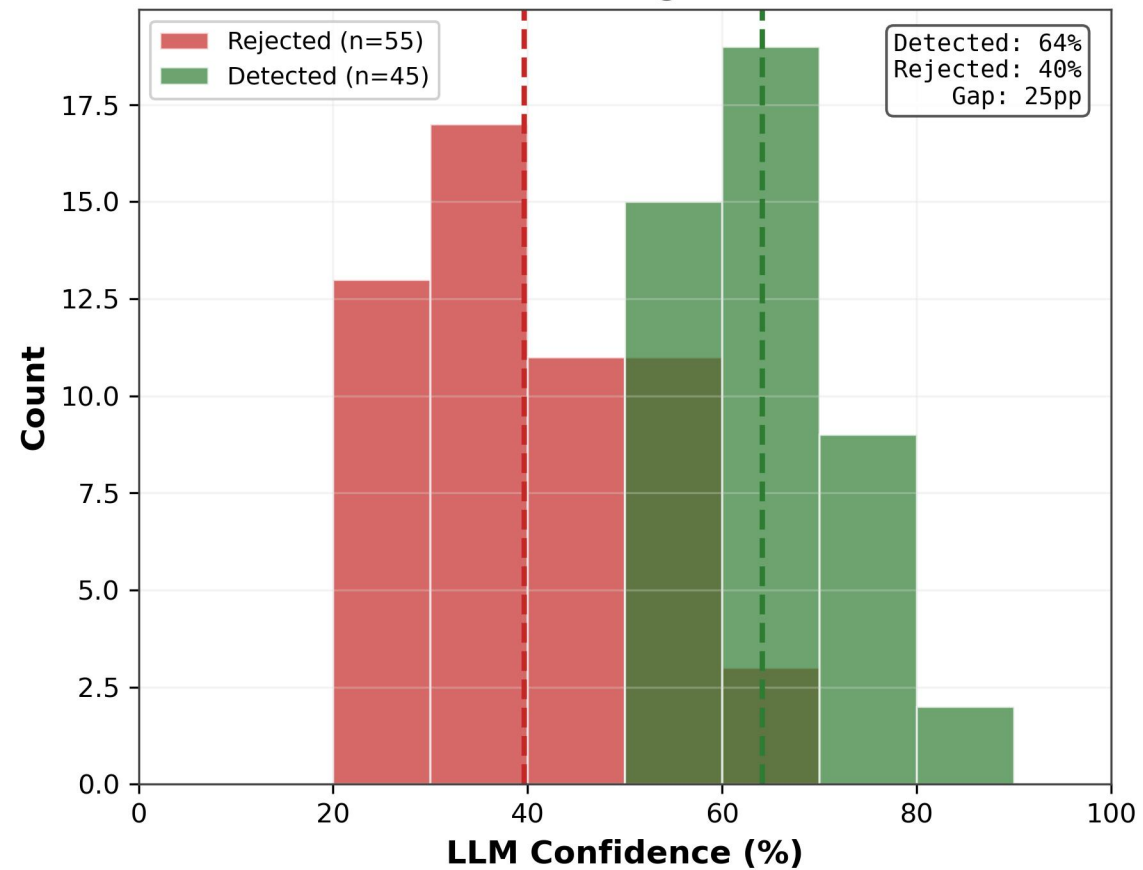
MEASUREMENT



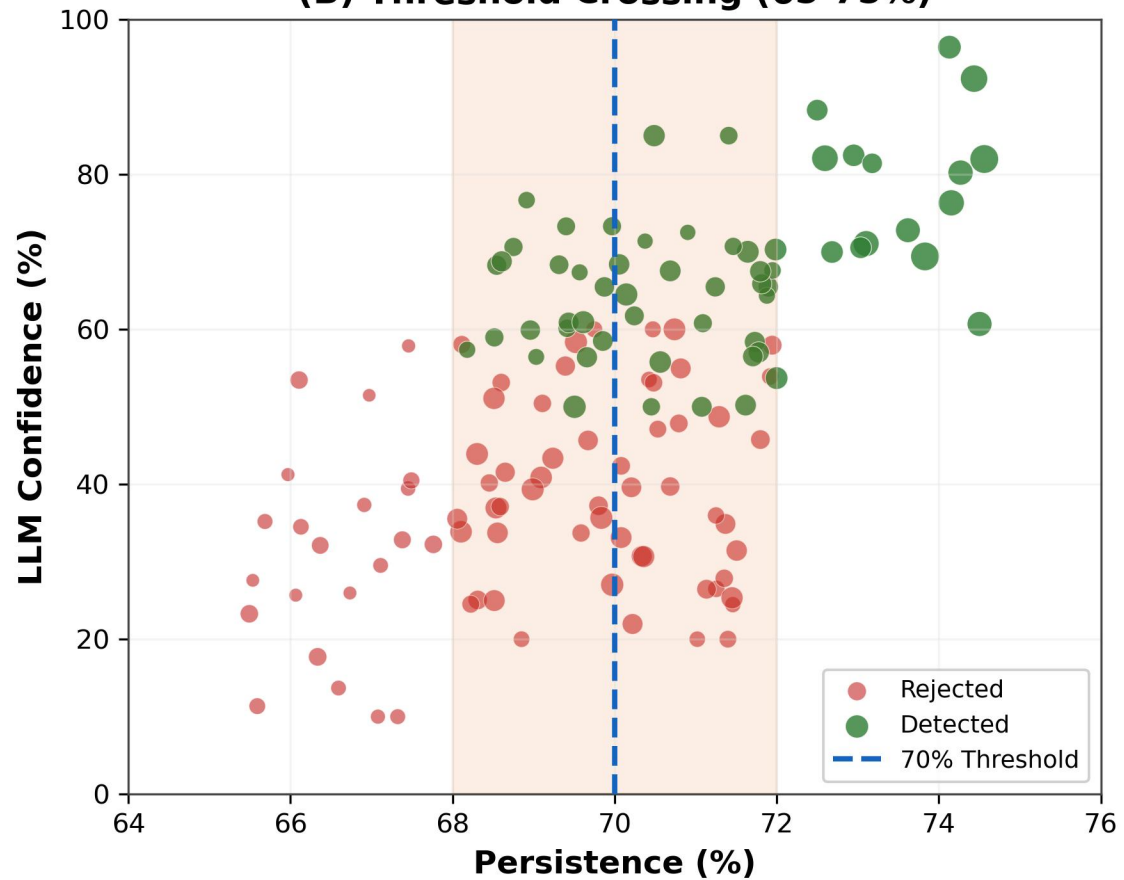
"Scar Tissue" Formation

Mechanism: High-frequency ODTE trading creates gamma exposure that dealers must hedge aggressively. When positions cannot be fully unwound by market close, the residual positioning is captured in EOD OI. This cumulative effect over time creates the persistent negative GEX regimes we detect.

(A) Borderline Region (68-72%)



(B) Threshold Crossing (65-75%)

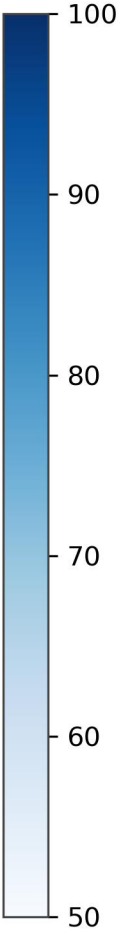


Current Parameters (70%, \$5B)

Persistence Threshold



Discrimination Gap (pp)
(2024% - 2020%)



All combinations >50pp: ✓ | Range: 82-99pp | Mean: 93pp